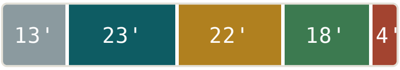
The projection of segments, and proportionality

Parallel lines cut any two transversals in the same ratios — the theorem of Thales, in full.

LESSON 53–54 | 2 × 40 MIN | GEOMETRIYA 9, §30 | IGX 11.2

LESSON CLOCK



Projection	13 min
Thales' theorem	23 min
Dividing a segment	22 min
The converse	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define the projection of a segment onto a line.
- State and use Thales' theorem for a family of parallel lines.
- Divide a segment in a given ratio by construction.
- Use the theorem to prove that a line is parallel to a side.

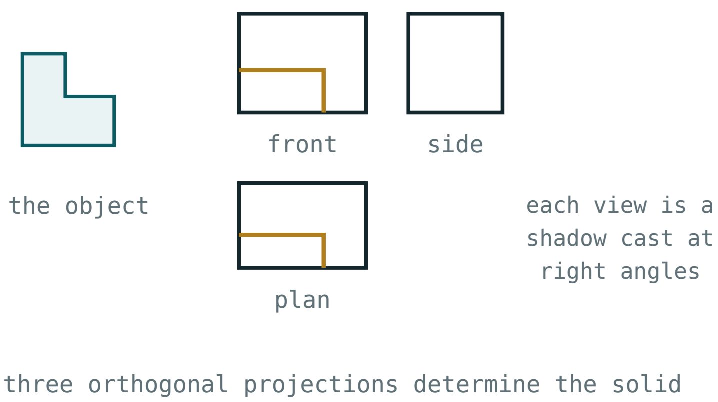
TEXTBOOK

National	pp. 183–189
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

Projection

The **projection** of a point onto a line is the foot of the perpendicular from it. The projection of a segment AB is the segment $A'B'$ joining the projections of its ends.



Dropping perpendiculars from both ends gives the projection of a segment.

SEGMENT	ITS PROJECTION
parallel to the line	equal in length
at α to the line	$AB \cos \alpha$
perpendicular to the line	a single point

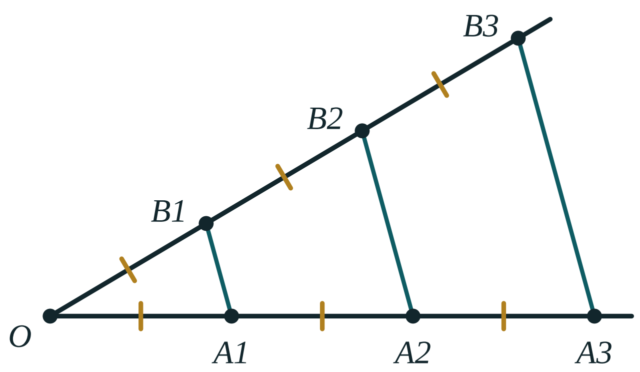
In a right triangle, the projections of the two legs onto the hypotenuse are the two parts into which the altitude divides it — the figure of Quarter I, met again.

A PROJECTION IS NEVER LONGER THAN THE SEGMENT

Because $\cos \alpha \leq 1$. That gives an immediate check on any answer, and it is the reason a shadow is shorter than the object except when the light is level.

Thales' theorem

> Parallel lines cutting two transversals cut off proportional segments on them



Three parallels across two transversals — the ratios on one match the ratios on the other.

If three parallels cut one transversal in A, B, C and the other in A_1, B_1, C_1, then

$$\frac{AB}{BC} = \frac{A_1B_1}{B_1C_1}$$

In particular, if the parallels cut equal segments on one transversal they cut equal segments on every transversal — which is the classical form of the theorem, and the basis of the construction in the next section.

THE TRANSVERSALS NEED NOT BE PARALLEL TO EACH OTHER

They may meet, cross, or be at any angle. Only the **cutting** lines must be parallel. Applying the theorem to a figure with non-parallel cutting lines is the error to watch for.

Dividing a segment

Problem. Divide a given segment AB in the ratio $3 : 5$ using only compasses and a straight edge.

STEP	WHAT TO DO
1	draw any ray AX from A
2	mark 8 equal steps along it: $P_1, \ldots, P_8$
3	join P_8 to B
4	draw the parallel to P_8B through P_3
5	it meets AB at the required point

The equal steps are made with one compass opening, and the parallels do the rest. No measuring is required, and the construction works for any ratio of whole numbers.

$3 : 5$ NEEDS 8 STEPS, NOT 5

The total of the two parts of the ratio is the number of equal steps to mark. Marking 5 and counting 3 gives $3 : 2$ instead.

The converse

> If a line divides two sides of a triangle proportionally, it is parallel to the third side

This is the form used in proofs. From $\frac{AD}{DB} = \frac{AE}{EC}$ it follows that $DE \parallel BC$ — and the second criterion of similarity is what proves it.

GIVEN	CONCLUDE
$DE \parallel BC$	$\dfrac{AD}{DB} = \dfrac{AE}{EC}$
$\dfrac{AD}{DB} = \dfrac{AE}{EC}$	$DE \parallel BC$
$DE \parallel BC$	$\dfrac{AD}{AB} = \dfrac{DE}{BC}$

TWO DIFFERENT RATIOS, BOTH CORRECT

$\dfrac{AD}{DB}$ uses the two **parts**; $\dfrac{AD}{AB}$ uses the part and the **whole**. Both are valid, but only the second gives $\dfrac{DE}{BC}$. Mixing them is the commonest error in this topic.

02 Worked examples

EXAMPLE 1 **In $\triangle ABC$, $DE \parallel BC$, $AD = 4$, $DB = 6$, $AE = 5$. Find EC.**

1. $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

   Parts to parts.

2. $\dfrac{4}{6} = \dfrac{5}{EC}$

3. $4 \cdot EC = 30$

4. $EC = 7.5$

ANSWER

$EC = 7.5$

EXAMPLE 2 **Three parallels cut one transversal into *3* and *7*. They cut the other into *4.5* and *x*. Find *x*.**

1. $\frac{3}{7} = \frac{4.5}{x}$
2. $3x = 31.5$
3. $x = 10.5$
4. Check: $\frac{4.5}{10.5} = \frac{3}{7}$ ✓

ANSWER

$x = 10.5$

EXAMPLE 3 **A segment *AB = 24* is divided in the ratio *3 : 5*. Find the two parts.**

1. Total parts: $3 + 5 = 8$.
2. One part = $\frac{24}{8} = 3$.
3. $3 \times 3 = 9$ and $5 \times 3 = 15$.
4. Check: $9 + 15 = 24$ ✓

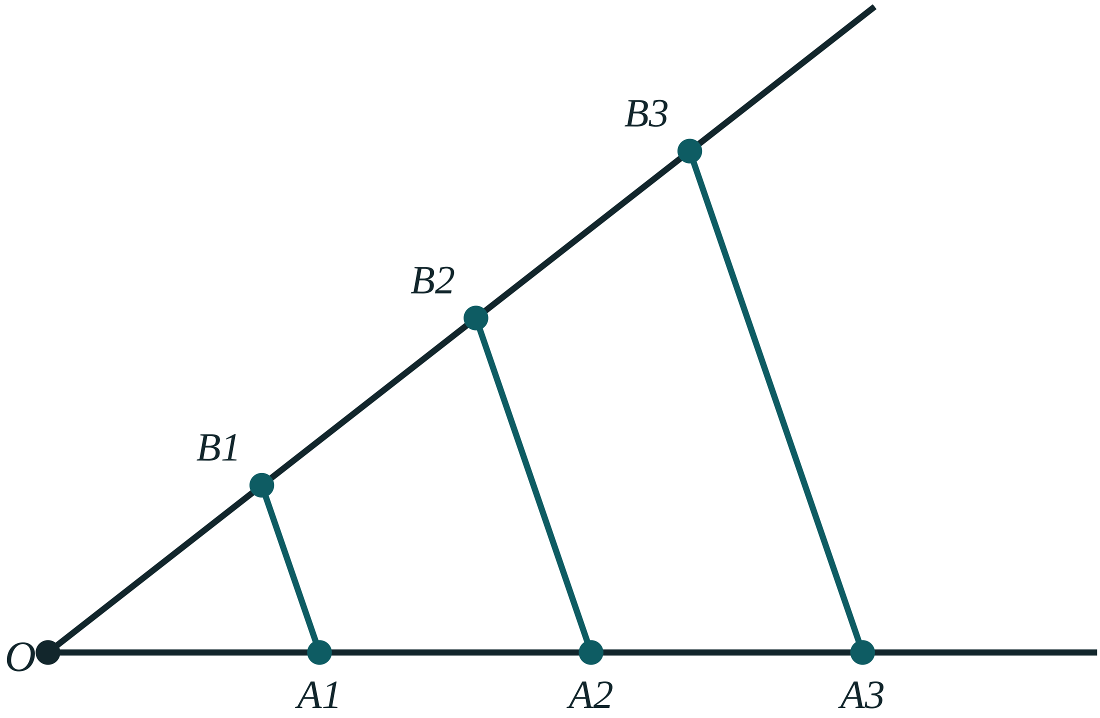
ANSWER

9 and 15

03 Interactive model

Divide a strip of paper into five equal parts using only a ruled exercise book: the printed parallel lines do the work, and Thales' theorem becomes a practical trick.

Thales' theorem



OA_1 **88**	A_1A_2 **88**	OB_1 **88**	B_1B_2 **88**	$OA_1 : OA_2$ **1 : 2**	$OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Projection	Proyeksiya	Проекция
Transversal	Kesuvchi	Секущая
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Thales’ theorem	Fales teoremasi	Теорема Фалеса
Proportional segments	Proporsional kesmalar	Пропорциональные отрезки
To divide in a ratio	Nisbatda bo‘lish	Разделить в отношении
Equal segments	Teng kesmalar	Равные отрезки
Converse	Teskari teorema	Обратная теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The projection of a segment at α to a line:

- AB
- $AB \cos \alpha$
- $AB \sin \alpha$
- $AB \tan \alpha$

Thales' theorem needs:

- parallel transversals
- parallel cutting lines
- a right angle
- equal segments

$DE \parallel BC$ gives:

- $\frac{AD}{DB} = \frac{AE}{EC}$
- $AD = AE$
- $DE = BC$
- nothing

$\frac{AD}{AB}$ equals:

- $\frac{DE}{BC}$
- $\frac{AD}{DB}$
- $\frac{BC}{DE}$
- 1

To divide in $3 : 5$, mark:

- 3
- 5
- 8
- 15

The converse of Thales gives:

- equal lengths
- parallelism
- a right angle
- similarity of areas

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $DE \parallel BC$, $AD = 4$, $DB = 6$, $AE = 5$: EC

   ANSWER 7.5

2. $AD = 3$, $DB = 6$, $AE = 4$: EC

   ANSWER 8

3. Divide 24 in $3 : 5$

   ANSWER 9 and 15

4. Divide 35 in $2 : 5$

   ANSWER 10 and 25

5. Projection of a segment parallel to the line

   ANSWER The segment itself

6. Projection of a perpendicular segment

   ANSWER A point

7. To divide in $2 : 7$, how many steps?

   ANSWER 9

Medium · the standard question

7 PROBLEMS

1. Parallels cutting 3, 7 and 4.5, x

   ANSWER $x = 10.5$

2. Parallels cutting 5, 8 and x, 12

   ANSWER $x = 7.5$

3. $DE \parallel BC$, $AD : AB = 2 : 5$: $DE : BC$

   ANSWER $2 : 5$

4. Same: $AD : DB$

   ANSWER $2 : 3$

5. Projection of a segment 10 at $60°$

   ANSWER 5

6. Projection of a segment 10 at $30°$

   ANSWER $5\sqrt{3}$

7. Is $DE \parallel BC$ if $AD = 4$, $DB = 6$, $AE = 6$, $EC = 9$?

   ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. A segment $AB = 30$ with $AP : PB = 2 : 3$: AP

   ANSWER 12

2. A triangle with a midline: the ratio it cuts each side in

   ANSWER $1 : 1$

3. Three parallels cut AB into x, $2x$, $3x$ and $CD = 24$: its parts

   ANSWER $4, 8, 12$

4. A trapezium: a line parallel to the bases through the diagonal intersection

   ANSWER Divides each leg equally by ratio

5. In $\triangle ABC$, D on AB with $AD : DB = 1 : 2$ and $DE \parallel BC$: $[\triangle ADE] : [ABC]$

   ANSWER $1 : 9$

6. Same figure: $[\triangle ADE] : [DBCE]$

   ANSWER $1 : 8$

7. A ladder against a wall makes $65°$: the projection of its $6\ m$ length on the ground

   ANSWER $\approx 2.54\ m$

07 Homework

Homework — 5 tasks

Say each time whether your ratio is part-to-part or part-to-whole.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 6$, $DB = 4$, $AE = 9$. Find EC.
2. Divide a segment of length 42 in the ratio $3 : 4$.
3. Three parallels cut one transversal into 4 and 9, and the other into 6 and x. Find x.
4. Describe the construction that divides a segment in the ratio $2 : 5$.
5. Find the projection of a segment of length 12 making $45°$ with a line.

Properties of proportional segments

The bisector theorems, internal and external — and the ratios they create.

LESSON 55–56 | 2 × 40 MIN | GEOMETRIYA 9, §31 | IGX 11.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The bisector theorem	13 min
Using it	23 min
Midlines and medians	22 min
Combining relations	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and prove the internal bisector theorem.
- Use it to find lengths and to find a side from a given division.
- Use the midline theorem and the median-intersection ratio.
- Combine several proportional relations in one problem.

TEXTBOOK

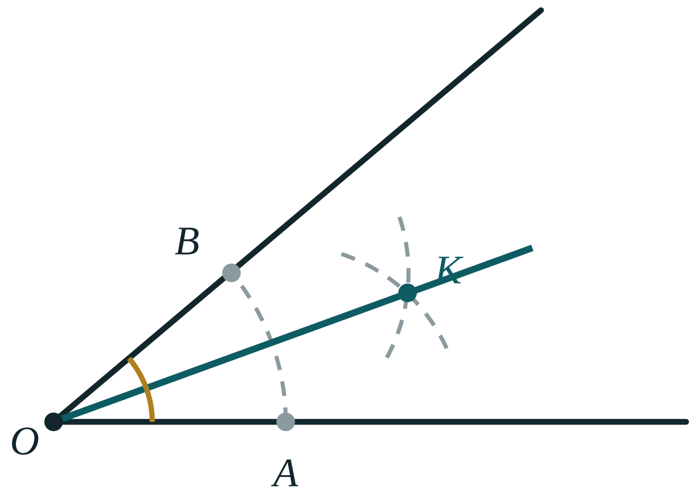
National	pp. 190–196
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

The bisector theorem

> The bisector of an angle of a triangle divides the opposite side in the ratio of the two adjacent sides

$$\frac{BD}{DC} = \frac{AB}{AC}$$



The bisector from A meets BC at D, dividing it as AB : AC.

Proof. Through C draw the parallel to AD, meeting BA extended at E. Then $\angle ACE = \angle CAD$ (alternate) and $\angle AEC = \angle BAD$ (corresponding), and since AD bisects, $\angle ACE = \angle AEC$, so $\triangle ACE$ is isosceles with $AE = AC$. Thales in $\triangle BEC$ then gives $\frac{\mathrm{BD}}{\mathrm{DC}} = \frac{\mathrm{BA}}{\mathrm{AE}} = \frac{\mathrm{AB}}{\mathrm{AC}}$.

THE BISECTOR IS THE ONLY CEVIAN WITH THIS PROPERTY

A median divides the opposite side in $1 : 1$ regardless of the sides; an altitude divides it in the ratio of the squares. Only the bisector reproduces the ratio of the sides themselves.

Using it

GIVEN	FIND	WORKING
$AB = 8, AC = 12, BC = 15$	BD	$\frac{2}{5} \times 15 = 6$
$AB = 10, BD = 4, DC = 6$	AC	15
$AB = 6, AC = 9, BD = 4$	DC	6
$AB = AC$	$BD : DC$	$1 : 1$

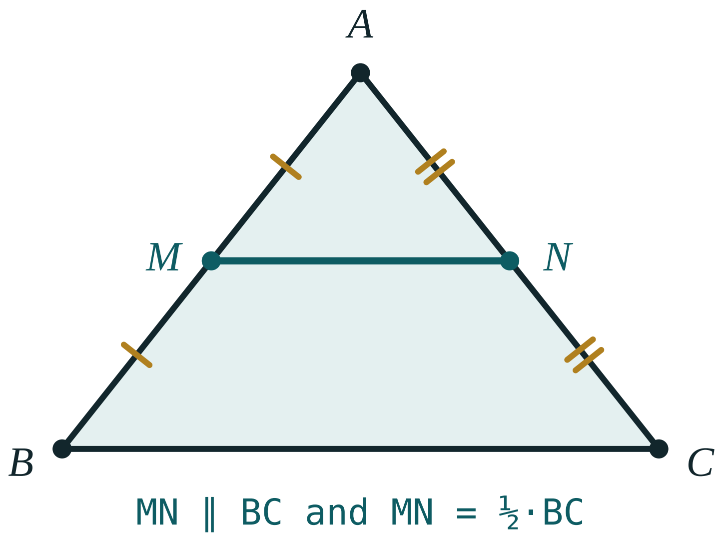
The last row says that in an isosceles triangle the bisector from the apex is also the median — and, by symmetry, the altitude.

SET UP THE PROPORTION BEFORE SUBSTITUTING

Write $\frac{BD}{DC} = \frac{AB}{AC}$ first, then put the numbers in. Half the errors in this topic come from pairing BD with AC.

Midlines and medians

STATEMENT	RATIO
the midline of a triangle	parallel to the third side and half of it
the midline of a trapezium	$\frac{a + b}{2}$
the three medians	meet at one point
the centroid divides each median	$2 : 1$



The midline joins two midpoints; it is parallel to the third side and half its length.

The $2 : 1$ ratio is measured **from the vertex**: the longer piece touches the vertex, the shorter the midpoint of the opposite side.

THE MIDLINE IS THALES WITH THE RATIO $1 : 1$

Everything in this section is one theorem seen at different ratios. Recognising that keeps four separate facts down to one.

Combining relations

A typical Quarter IV problem chains two or three of these together.

Example. In $\triangle ABC$, $AB = 12$, $AC = 18$, $BC = 20$. The bisector from A meets BC at D, and M is the midpoint of BC. Find DM.

STEP	WORKING
$BD : DC$	$12 : 18 = 2 : 3$
BD	$\frac{2}{5} \times 20 = 8$
BM	10
DM	$10 - 8 = 2$

PUT EVERY COMPUTED LENGTH ON THE DIAGRAM

A figure with $BD = 8$ and $BM = 10$ marked answers the question at a glance. Keeping the numbers in the working instead of on the picture is what makes these problems feel hard.

02 Worked examples

EXAMPLE 1 **In $\triangle ABC$, $AB = 8$, $AC = 12$, $BC = 15$. The bisector from A meets BC at D. Find BD and DC.**

1. $$\frac{\mathrm{BD}}{\mathrm{DC}} = \frac{8}{12} = \frac{2}{3}$$
2. Total parts 5.
3. $BD = \frac{2}{5} \times 15 = 6$
4. $DC = 9$

ANSWER

$BD = 6, DC = 9$

EXAMPLE 2 **In $\triangle ABC$, $AB = 10$ and the bisector from A divides BC into 4 and 6. Find AC.**

1. $$\frac{4}{6} = \frac{10}{\mathrm{AC}}$$
2. $4 \cdot AC = 60$
3. $AC = 15$
4. Check: $\frac{10}{15} = \frac{2}{3} = \frac{4}{6}$ ✓

ANSWER

$AC = 15$

EXAMPLE 3 **In $\triangle ABC$, $AB = 12$, $AC = 18$, $BC = 20$. Find the distance from the foot of the bisector from A to the midpoint of BC.**

1. $BD : DC = 12 : 18 = 2 : 3$
2. $BD = \frac{2}{5} \times 20 = 8$
3. $BM = 10$

   The midpoint.
4. $DM = 2$

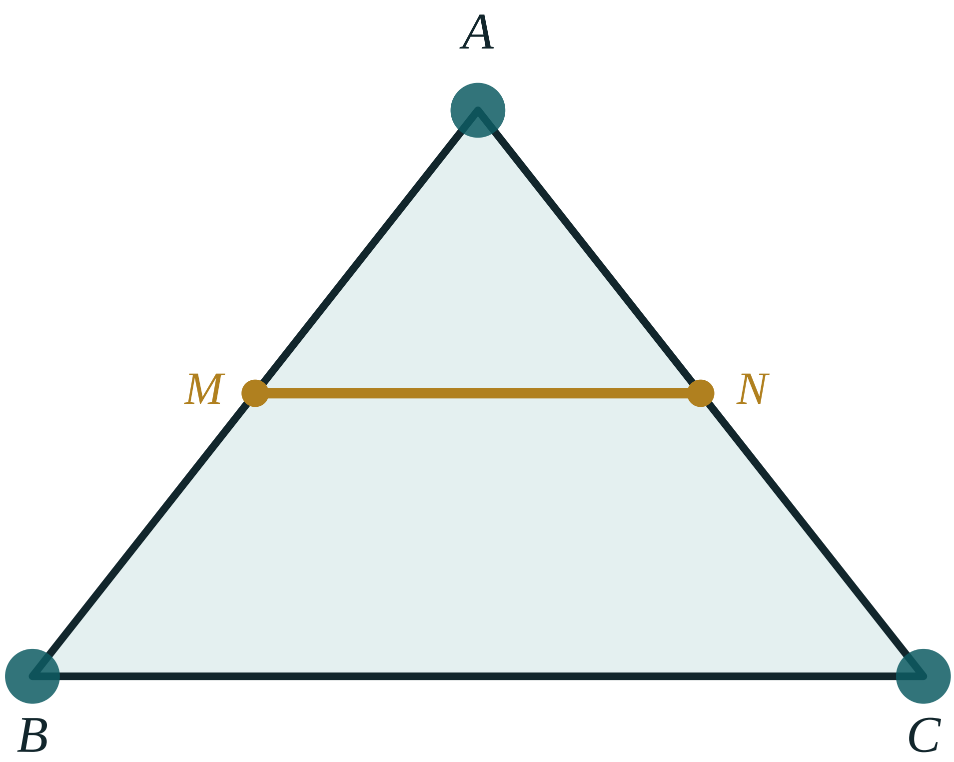
ANSWER

$DM = 2$

03 Interactive model

Draw a triangle with sides 3 cm and 6 cm and bisect the angle between them; measuring the two parts of the opposite side confirms 1 : 2 before any proof.

The midline of a triangle



BC **260** | MN **130** | BC / MN **2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Bisector theorem	Bissektrisa teoremasi	Теорема о биссектрисе
Internal bisector	Ichki bissektrisa	Внутренняя биссектриса
Midline	O‘rta chiziq	Средняя линия
Median	Mediana	Медиана
Centroid	Og‘irlik markazi	Центр тяжести
Ratio	Nisbat	Отношение
Proportional	Proporsional	Пропорциональный
Auxiliary parallel	Yordamchi parallel	Вспомогательная параллель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The bisector from A divides BC in the ratio:

- $AB : BC$
- $AB : AC$
- $AC : BC$
- $1 : 1$

$AB = 8$, $AC = 12$, $BC = 15$: $BD =$

- 5
- 6
- 7.5
- 9

In an isosceles triangle the bisector from the apex is also:

- an altitude only
- a median only
- both
- neither

A midline is:

- equal to the third side
- half the third side
- twice it
- unrelated

The centroid divides each median in:

- $1 : 1$
- $2 : 1$
- $3 : 1$
- $1 : 2$ from the vertex

The midline of a trapezium is:

- $a + b$
- $\frac{a + b}{2}$
- $\frac{ab}{2}$
- $a - b$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $AB = 8, AC = 12$: $BD : DC$

   ANSWER $2 : 3$

2. $AB = 8, AC = 12, BC = 15$: BD

   ANSWER 6

3. Same: DC

   ANSWER 9

4. $AB = 6, AC = 9, BD = 4$: DC

   ANSWER 6

5. $AB = AC$: $BD : DC$

   ANSWER $1 : 1$

6. Midline of a triangle with third side 14

   ANSWER 7

7. The centroid ratio

   ANSWER $2 : 1$

Medium · the standard question

7 PROBLEMS

1. $AB = 10$, $BD = 4$, $DC = 6$: AC

ANSWER 15

2. $AB = 12$, $AC = 18$, $BC = 20$: BD

ANSWER 8

3. Same: DM where M is the midpoint

ANSWER 2

4. Midline of a trapezium with bases 8 and 14

ANSWER 11

5. A median of length 12: the distance from the vertex to the centroid

ANSWER 8

6. $AB = 15$, $AC = 20$, $BC = 21$: BD

ANSWER 9

7. A triangle with $BD : DC = 3 : 4$ and $AB = 9$: AC

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. A triangle $13, 14, 15$: the bisector from the vertex between 13 and 14 divides 15

   ANSWER 7.22 and 7.78

2. A trapezium with bases 10 and 16: the segment of the midline between the diagonals

   ANSWER 3

3. A triangle whose medians are $9, 12, 15$: its area

   ANSWER 72

4. A bisector divides $BC = 28$ as $3 : 4$ with $AB = 12$: AC

   ANSWER 16

5. The centroid of a triangle with vertices $(0,0), (6,0), (0,9)$

   ANSWER $(2, 3)$

6. A median divides a triangle into two parts of area ratio

   ANSWER $1 : 1$

7. Three medians divide a triangle into

   ANSWER 6 triangles of equal area

07 Homework

Homework — 5 tasks

Write the proportion in full before substituting any number.

1. In $\triangle ABC$, $AB = 9$, $AC = 15$, $BC = 16$. Find the parts into which the bisector from A divides BC.
2. A bisector divides BC into 5 and 7, and $AB = 20$. Find AC.
3. Find the midline of a trapezium with bases 12 and 20.
4. A median has length 15. Find the two parts into which the centroid divides it.
5. Prove the bisector theorem using an auxiliary parallel.

Proportional segments in a right-angled triangle

The altitude to the hypotenuse, and the three mean proportionals it creates.

LESSON 57–58 | 2 × 40 MIN | GEOMETRIYA 9, §32 | IGX 11.1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The mean proportional	13 min
The three relations	23 min
Pythagoras from them	22 min
Problems	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define the mean proportional of two segments.
- State the three relations of the altitude figure and use them fluently.
- Derive Pythagoras' theorem from them.
- Solve numerical problems on the altitude and the projections.

TEXTBOOK

National	pp. 197–203
Cambridge	Core & Extended, pp. 220–226
Workbook	Exercise 11.1

01 Explanation

The mean proportional

x is the **mean proportional** between a and b if

$$\frac{a}{x} = \frac{x}{b} \text{ that is } x^2 = ab \text{ that is } x = \sqrt{ab}$$

It is the geometric mean of the algebra course, met here as a length. Since a length is positive, only the positive root is taken.

A	B	$X = \sqrt{AB}$
4	9	6
2	8	4
3	12	6
5	20	10

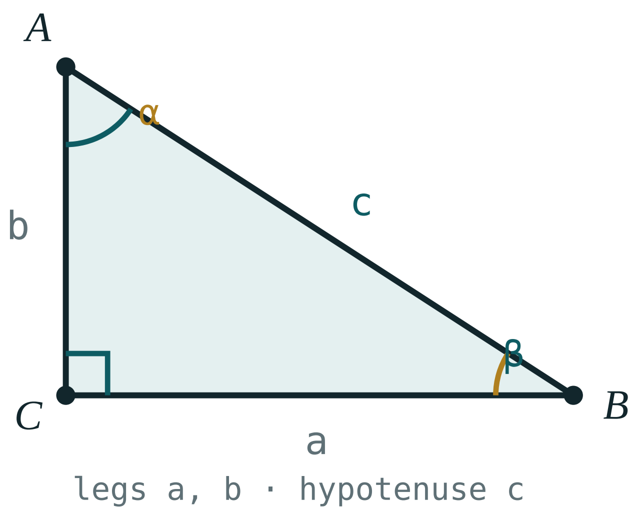
THE MEAN PROPORTIONAL ALWAYS LIES BETWEEN THE TWO

Because $\sqrt{ab}$ is between a and b whenever they differ — the arithmetic–geometric mean inequality of the algebra course, seen as a picture.

The three relations

In a right triangle with the right angle at C and altitude CH to the hypotenuse AB:

$$CH^2 = AH \cdot HB \quad AC^2 = AH \cdot AB \quad BC^2 = BH \cdot AB$$



The altitude divides the triangle into two triangles similar to it and to each other.

SEGMENT	IS THE MEAN PROPORTIONAL BETWEEN
the altitude CH	the two projections
the leg AC	its own projection and the whole hypotenuse
the leg BC	its own projection and the whole hypotenuse

All three come from the similarity of the three triangles, and each is one cross-multiplication away from a proportion.

EACH LEG GOES WITH ITS *OWN* PROJECTION

AC with AH, BC with BH. Crossing them gives a plausible number and a wrong answer, and it is by far the commonest error of this lesson.

Pythagoras from them

Add the two leg relations:

$$AC^2 + BC^2 = AH \cdot AB + BH \cdot AB = (AH + BH) \cdot AB = AB \cdot AB = AB^2$$

That is Pythagoras' theorem, in one line, from similar triangles alone — a proof quite different from the area proofs of Grade 8.

ALSO WORTH KNOWING	STATEMENT
the altitude	$CH = \dfrac{AC \cdot BC}{AB}$
the area, two ways	$\tfrac{1}{2} \cdot AC \cdot BC = \tfrac{1}{2} \cdot AB \cdot CH$
the reciprocal relation	$\dfrac{1}{CH^2} = \dfrac{1}{AC^2} + \dfrac{1}{BC^2}$

TWO EXPRESSIONS FOR THE SAME AREA

The middle row is where $CH = \dfrac{AC \cdot BC}{AB}$ comes from, and it is often the fastest route to the altitude when both legs are known.

Problems

GIVEN	FIND	RELATION
projections 9 and 16	CH	$\sqrt{144} = 12$
same	AC, BC	$\sqrt{9 \cdot 25} = 15, \sqrt{16 \cdot 25} = 20$
legs 6 and 8	CH	$\dfrac{48}{10} = 4.8$
same	the projections	$3.6 \text{ and } 6.4$
$AB = 25, AC = 15$	AH	$\dfrac{225}{25} = 9$

LABEL THE FIGURE BEFORE WRITING ANY RELATION

Mark A, B, C, H and every known length. The three relations are unusable on an unlabelled diagram, and readable at a glance on a labelled one.

02 Worked examples

EXAMPLE 1 **The altitude to the hypotenuse divides it into 9 and 16. Find the altitude and the two legs.**

1. $CH^2 = 9 \times 16 = 144 \Rightarrow CH = 12$
2. $AB = 25$
3. $AC^2 = 9 \times 25 = 225 \Rightarrow AC = 15$
4. $BC^2 = 16 \times 25 = 400 \Rightarrow BC = 20$

   Check: $15^2 + 20^2 = 625$ ✓

ANSWER

$CH = 12, AC = 15, BC = 20$

EXAMPLE 2 **A right triangle has legs *6* and *8*. Find the altitude to the hypotenuse and the projections.**

1. $AB = 10$

2. $CH = \dfrac{6 \times 8}{10} = 4.8$

   Two areas.

3. $AH = \dfrac{36}{10} = 3.6$

4. $BH = \dfrac{64}{10} = 6.4$

   Check: $3.6 + 6.4 = 10$ ✓

ANSWER

$CH = 4.8, AH = 3.6, BH = 6.4$

EXAMPLE 3 **Find the mean proportional between *3* and *27*, and between *5* and *20*.**

1. $x = \sqrt{3 \times 27} = \sqrt{81}$

2. $= 9$

3. $x = \sqrt{5 \times 20} = \sqrt{100}$

4. $= 10$

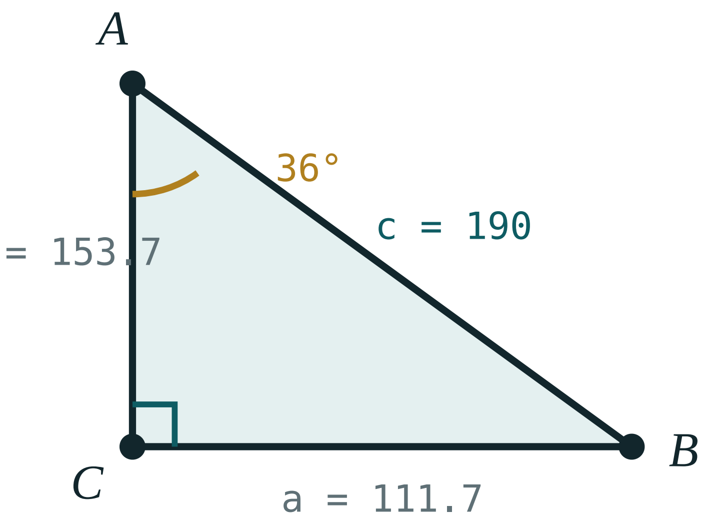
ANSWER

9 and 10

03 Interactive model

Draw the altitude figure once, very large, and label all six lengths; every question in the lesson is then read off that one picture.

The altitude figure



α **36°** · β **54°** · a (opposite) **111.7** · b (adjacent) **153.7** · c (hyp) **190** · sin α **0.59**

cos α **0.81** · tan α **0.73** · $a^2 + b^2$ **36100** · c^2 **36100**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mean proportional	O‘rta proporsional	Среднее пропорциональное
Geometric mean	O‘rta geometrik	Среднее геометрическое
Altitude	Balandlik	Высота
Projection	Proyeksiya	Проекция
Hypotenuse	Gipotenuza	Гипотенуза
Leg	Katet	Катет
Relation	Munosabat	Соотношение
Derive	Keltirib chiqarish	Вывести

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The mean proportional between a and b:

- $\frac{a + b}{2}$
- $\sqrt{ab}$
- ab
- $a - b$

CH^2 equals:

- $AH \cdot AB$
- $AH \cdot HB$
- $HB \cdot AB$
- AB^2

AC^2 equals:

- $AH \cdot HB$
- $AH \cdot AB$
- $HB \cdot AB$
- CH^2

Adding the two leg relations gives:

- the sine rule
- Pythagoras
- the area
- nothing

Legs 6 and 8: the altitude is

- 4
- 4.8
- 5
- 7

The mean proportional between 4 and 9:

- 6
- 6.5
- 13
- 36

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Mean proportional of 4 and 9

ANSWER 6

2. Mean proportional of 2 and 8

ANSWER 4

3. Mean proportional of 5 and 20

ANSWER 10

4. Projections 9 and 16: the altitude

ANSWER 12

5. Same: the hypotenuse

ANSWER 25

6. Same: the legs

ANSWER 15 and 20

7. Legs 6 and 8: the altitude

ANSWER 4.8

Medium · the standard question

7 PROBLEMS

1. Legs 6 and 8: the projections

ANSWER 3.6 and 6.4

2. Legs 5 and 12: the altitude

ANSWER $\frac{60}{13}$

3. $AB = 25, AC = 15$: AH

ANSWER 9

4. $AB = 25, AC = 15$: CH

ANSWER 12

5. Projections 4 and 12: the altitude

ANSWER $4\sqrt{3}$

6. Altitude 6, one projection 4: the other

ANSWER 9

7. Same: the hypotenuse

ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. A right triangle with hypotenuse 20 and altitude 8: the projections

   ANSWER 4 and 16

2. Legs a and b: the altitude

   ANSWER $\dfrac{ab}{\sqrt{a^2 + b^2}}$

3. Prove $\dfrac{1}{CH^2} = \dfrac{1}{AC^2} + \dfrac{1}{BC^2}$

   ANSWER Substitute $CH = \dfrac{ab}{c}$

4. A right triangle whose altitude divides the hypotenuse in $1 : 4$: the ratio of the legs

   ANSWER $1 : 2$

5. A right triangle with projections in ratio $9 : 16$: the ratio of the legs

   ANSWER $3 : 4$

6. The altitude of a right triangle with hypotenuse c equals $\dfrac{c}{2}$ when

   ANSWER The triangle is isosceles

7. A right triangle with altitude 13 and hypotenuse 25: possible?

   ANSWER No — the maximum altitude is 12.5

07 Homework

Homework — 5 tasks

Pair each leg with its own projection; write the pairing down before substituting.

1. The altitude to the hypotenuse divides it into 4 and 21. Find the altitude and the legs.
2. A right triangle has legs 9 and 12. Find the altitude and the projections.
3. Find the mean proportional between 8 and 18.

4. A right triangle has hypotenuse 26 and one leg 10. Find that leg's projection.
5. Derive Pythagoras' theorem from the two leg relations.

Constructing the mean proportional of two segments

Compasses and a straight edge produce √ab exactly — no measuring, no decimals.

LESSON 59–60 | 2 × 40 MIN | GEOMETRIYA 9, §33 | IGX 3.6

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The construction	13 min
Why it works	23 min
Constructing surds	22 min
Squaring a rectangle	18 min
Homework	4 min

LEARNING OBJECTIVES

- Carry out the semicircle construction for the mean proportional.
- Justify it with the altitude relation.
- Construct √a for a given segment a, and hence √2, √3, √5.
- Use the construction to build a square equal in area to a given rectangle.

TEXTBOOK

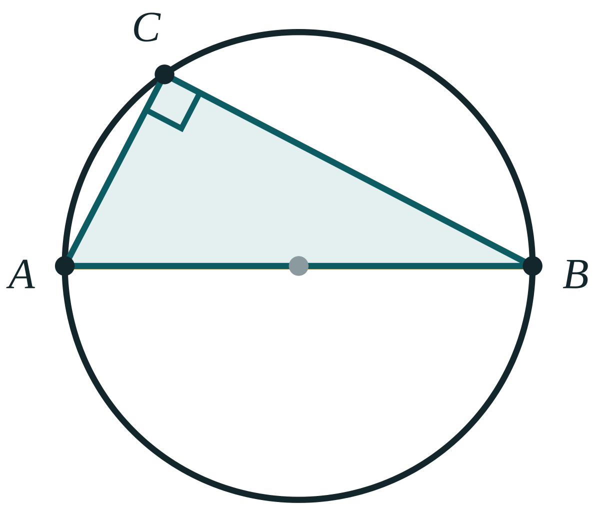
National	pp. 204–209
Cambridge	Core & Extended, pp. 70–75
Workbook	Exercise 3.6

01 Explanation

The construction

Problem. Given segments of lengths a and b, construct a segment of length $\sqrt{ab}$.

STEP	WHAT TO DO
1	on a line, mark $AH = a$ and then $HB = b$, in that order
2	find the midpoint of AB with the perpendicular-bisector construction
3	draw the semicircle on AB as diameter
4	erect the perpendicular to AB at H
5	it meets the semicircle at C; then $CH = \sqrt{ab}$



Any point on a semicircle sees the diameter at a right angle.

THE WHOLE CONSTRUCTION IS THALES' SEMICIRCLE THEOREM

Because C lies on the semicircle, $\angle ACB = 90°$. So CH is the altitude of a right triangle to its hypotenuse, and the relation of the last lesson applies.

Why it works

In the figure, $\triangle ACB$ is right-angled at C and $CH \perp AB$. Therefore

$$CH^2 = AH \cdot HB = ab \text{ so } CH = \sqrt{ab}$$

The construction is exact — not an approximation. That is the point of a straight-edge and compasses construction: it produces the number itself, however irrational.

A	B	CH
1	2	$\sqrt{2}$
1	3	$\sqrt{3}$
1	5	$\sqrt{5}$
2	8	4

A AND *B* MUST BE LAID END TO END

$AH = a$ and $HB = b$ along the **same** line, so that $AB = a + b$ is the diameter. Marking them from the same end gives the wrong figure entirely.

Constructing surds

Taking $a = 1$ makes $CH = \sqrt{b}$. So from a unit segment and a segment of length b, the construction produces $\sqrt{b}$ exactly.

TO CONSTRUCT	TAKE
$\sqrt{2}$	$a = 1, b = 2$
$\sqrt{3}$	$a = 1, b = 3$
$\sqrt{6}$	$a = 2, b = 3$
$\sqrt{10}$	$a = 2, b = 5$

There is a second route for $\sqrt{2}$: the diagonal of a unit square. For $\sqrt{3}$: the diagonal of a unit cube, or the height of an equilateral triangle of side 2. The semicircle construction is the one that works for every b.

NOT EVERY NUMBER CAN BE CONSTRUCTED

Square roots can; $\sqrt[3]{2}$ cannot, and neither can π. Proving that took until the nineteenth century, and it is why “squaring the circle” is impossible — a phrase that entered ordinary language from this very construction.

Squaring a rectangle

Problem. Construct a square with the same area as a given rectangle $a \times b$.

The square must have side x with $x^2 = ab$ — that is, $x = \sqrt{ab}$, which is exactly what the construction produces.

RECTANGLE	EQUAL SQUARE
4×9	side 6
2×8	side 4
3×5	side $\sqrt{15}$
$a \times b$	side $\sqrt{ab}$

THIS IS WHAT "QUADRATURE" MEANT TO THE GREEKS

To find the area of a figure was to construct a square equal to it. Rectangles, triangles and polygons can all be squared; the circle cannot, and that question stayed open for two thousand years.

02 Worked examples

EXAMPLE 1 **Describe the construction of $\sqrt{12}$ from a unit segment.**

1. Take $a = 1$ and $b = 12$ end to end.
   Or $a = 3$, $b = 4$.
2. Draw the semicircle on the total, 13, as diameter.
3. Erect the perpendicular at the join.
4. Its length to the semicircle is $\sqrt{12} = 2\sqrt{3}$.

ANSWER

$\sqrt{12} = 2\sqrt{3}$

EXAMPLE 2 **A rectangle is $3\ cm$ by $12\ cm$. Find the side of the square of equal area.**

1. $x^2 = 3 \times 12$
2. $= 36$
3. $x = 6$
4. Check: $6^2 = 36 = 3 \times 12$ ✓

ANSWER

$6\ cm$

EXAMPLE 3 **The perpendicular at H meets the semicircle at height $CH = 8$, and $AH = 4$. Find HB and the diameter.**

1. $CH^2 = AH \cdot HB$
2. $64 = 4 \cdot HB$
3. $HB = 16$
4. $AB = 4 + 16 = 20$

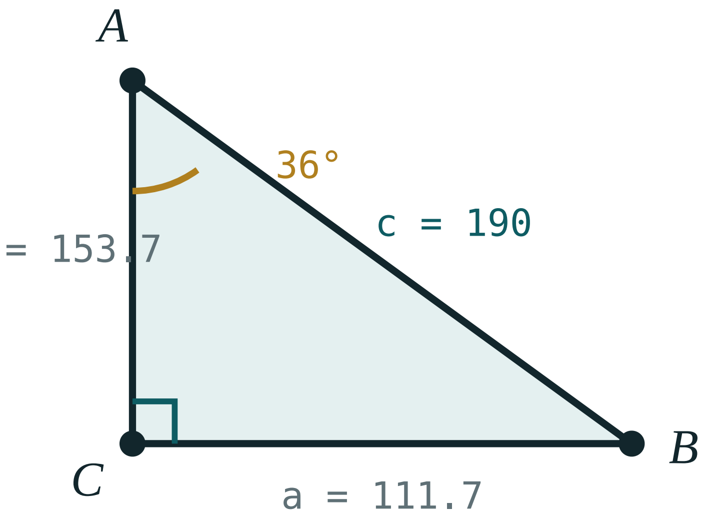
ANSWER

$HB = 16, AB = 20$

03 Interactive model

Do the construction on the board with real compasses for a = 1 cm and b = 3 cm, then measure CH; it comes out at 1.7 cm, and √3 = 1.732.

The semicircle construction



α **36°**	β **54°**	a (opposite) **111.7**	b (adjacent) **153.7**	c (hyp) **190**	sin α **0.59**
cos α **0.81**	tan α **0.73**	$a^2 + b^2$ **36100**	c^2 **36100**		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Compasses	Sirkul	Циркуль
Straight edge	Chizg‘ich	Линейка
Semicircle	Yarim aylana	Полуокружность
Perpendicular	Perpendikulyar	Перпендикуляр
Justification	Asoslash	Обоснование
Unit segment	Birlik kesma	Единичный отрезок
Quadrature	Kvadratura	Квадратура

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The construction rests on:

- the sine rule
- Thales' semicircle theorem
- Pythagoras only
- the bisector theorem

CH equals:

- $a + b$
- $\sqrt{ab}$
- $\frac{a + b}{2}$
- ab

a and b are laid:

- from the same end
- end to end
- perpendicular
- anywhere

To construct $\sqrt{5}$ take:

- $a = 1, b = 5$
- $a = 5, b = 5$
- $a = 2, b = 3$
- $a = 1, b = 25$

A square equal to a 3×12 rectangle has side:

- 6
- 7.5
- 15
- 36

Which cannot be constructed?

- $\sqrt{2}$
- $\sqrt{7}$
- π
- $\sqrt{15}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Mean proportional of 1 and 4

   ANSWER 2

2. Mean proportional of 1 and 9

   ANSWER 3

3. To construct $\sqrt{2}$ take

   ANSWER $a = 1, b = 2$

4. To construct $\sqrt{3}$ take

   ANSWER $a = 1, b = 3$

5. Square equal to a 4×9 rectangle

   ANSWER Side 6

6. Square equal to a 2×8 rectangle

   ANSWER Side 4

7. CH^2 equals

   ANSWER $AH \cdot HB$

Medium · the standard question

7 PROBLEMS

1. Square equal to a 3×12 rectangle

ANSWER Side 6

2. Square equal to a 3×5 rectangle

ANSWER Side $\sqrt{15}$

3. $CH = 8, AH = 4$: HB

ANSWER 16

4. Same: the diameter

ANSWER 20

5. To construct $\sqrt{6}$ take

ANSWER $a = 2, b = 3$

6. To construct $\sqrt{10}$ take

ANSWER $a = 2, b = 5$

7. $AH = 2, HB = 18$: CH

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. Construct $\sqrt{12}$: one choice of a, b

ANSWER 3 and 4

2. A rectangle of perimeter 26 and area 36: the equal square's side

ANSWER 6

3. The greatest CH for a fixed $AB = c$

ANSWER $\frac{c}{2}$, when H is the midpoint

4. Which is larger, $\frac{a + b}{2}$ or $\sqrt{ab}$?

ANSWER The arithmetic mean, unless $a = b$

5. Interpret that inequality in the semicircle

ANSWER The radius is at least the altitude

6. Construct a square equal to a triangle of base 8 and height 6

ANSWER Side $\sqrt{24} = 2\sqrt{6}$

7. Why can a circle not be squared?

ANSWER π is not constructible

07 Homework

Homework — 5 tasks

Describe every construction as numbered steps, and justify it in one sentence.

1. Describe the construction of the mean proportional between $2\ cm$ and $8\ cm$.
2. Find the side of the square equal in area to a 5×20 rectangle.
3. Describe how to construct $\sqrt{7}$ from a unit segment.
4. In the semicircle figure, $AH = 3$ and $CH = 6$. Find HB and the diameter.

5. Explain why the greatest possible altitude in the figure is half the diameter.

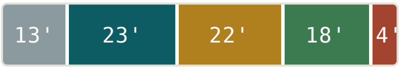
Home / Grade 9 / Geometry / Quarter 4

Proportional segments in a circle

Chords, secants and a tangent — three theorems that are one theorem.

LESSON 61–62 | 2 × 40 MIN | GEOMETRIYA 9, §34 | IGX 3.5

LESSON CLOCK



Two chords inside	13 min
Two secants outside	23 min
A tangent and a secant	22 min
One theorem	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and prove the intersecting-chords theorem.
- State and use the secant–secant and tangent–secant theorems.
- Recognise that all three are the power of a point.
- Solve numerical problems on each configuration.

TEXTBOOK

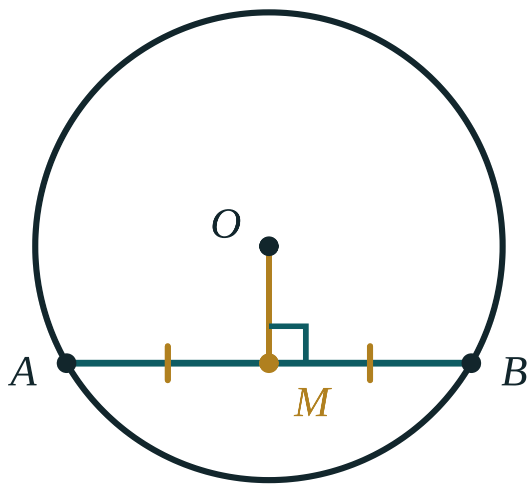
National	pp. 210–216
Cambridge	Core & Extended, pp. 63–70
Workbook	Exercise 3.5

01 Explanation

Two chords inside

> If two chords meet inside a circle, the products of the two parts of each are equal

$$PA \cdot PB = PC \cdot PD$$



$$OM \perp AB \implies AM = MB$$

Two chords crossing at P — the two products are equal.

Proof. Join A to C and D to B. Then $\angle APC = \angle DPB$ (vertical) and $\angle CAP = \angle BDP$ (angles on the arc CB), so $\triangle APC \sim \triangle DPB$. Hence $\dfrac{PA}{PD} = \dfrac{PC}{PB}$, and cross-multiplying gives the result.

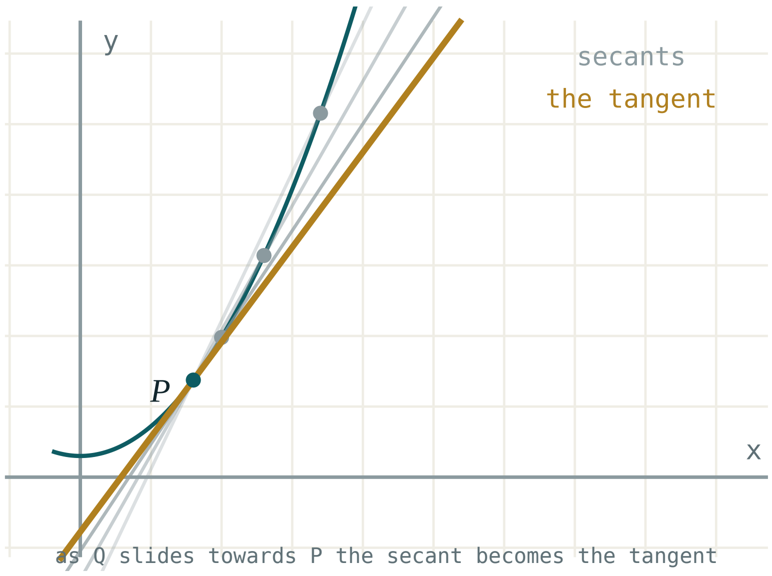
THE PROOF IS THE WHOLE CHAPTER IN MINIATURE

Two equal angles, a similarity, a proportion, a cross-multiplication. Every theorem in this section follows the same four steps with a different pair of triangles.

Two secants outside

From a point P outside the circle, draw two secants meeting it at A, B and C, D (with A and C nearer to P). Then

$$PA \cdot PB = PC \cdot PD$$



Two secants from an external point — again the products agree.

The proof is the same: $\angle P$ is common and $\angle PAC = \angle PDB$ because $ABDC$ is cyclic, so $\triangle PAC \sim \triangle PDB$.

> THE PRODUCTS USE THE *WHOLE* SECANT, NOT THE FAR PART
>
> PB runs from P all the way to the far intersection, not from A to B. Using AB in place of PB is the standard error, and it gives an answer that looks reasonable.

A tangent and a secant

If PT is a tangent and PAB a secant from the same external point,

$$PT^2 = PA \cdot PB$$

so the tangent length is the mean proportional between the two parts of the secant — the construction of the last lesson, appearing in a circle.

GIVEN	FIND	WORKING
$PA = 4, PB = 9$	PT	$\sqrt{36} = 6$
$PT = 8, PA = 4$	PB	16
$PT = 12, PB = 18$	PA	8
$PA = 5, AB = 11$	PT	$\sqrt{80} = 4\sqrt{5}$

A TANGENT IS THE LIMITING CASE OF A SECANT

Rotate a secant until its two intersections merge into one: then PA and PB both become PT, and $PA \cdot PB$ becomes PT^2. The third theorem is the second one at its limit.

One theorem

Define the **power** of a point P with respect to a circle of centre O and radius R as

$$k = OP^2 - R^2$$

POSITION OF P	K	EQUALS
outside	> 0	$PA \cdot PB = PT^2$
on the circle	0	0
inside	< 0	$-PA \cdot PB$

So all three theorems say the same thing: the product $PA \cdot PB$ depends only on P and the circle, not on which line through P is drawn.

THREE THEOREMS, ONE IDEA, ONE THING TO REMEMBER

Draw any line through P; multiply the two distances to the circle. The answer is always the same. That sentence replaces all three statements.

02 Worked examples

EXAMPLE 1 **Two chords meet at P with $PA = 4, PB = 9, PC = 6$. Find PD.**

1. $PA \cdot PB = PC \cdot PD$
2. $4 \times 9 = 6 \times PD$
3. $36 = 6\ PD$
4. $PD = 6$

ANSWER

$PD = 6$

EXAMPLE 2 **From P, a tangent PT and a secant meeting the circle at A and B with $PA = 4$ and $AB = 5$. Find PT.**

1. $PB = PA + AB = 9$

   The whole secant.
2. $PT^2 = PA \cdot PB = 4 \times 9$
3. $= 36$
4. $PT = 6$

ANSWER

$PT = 6$

EXAMPLE 3 **From an external point, two secants give $PA = 5$, $PB = 12$ and $PC = 4$. Find PD.**

1. $PA \cdot PB = PC \cdot PD$
2. $5 \times 12 = 4 \times PD$
3. $60 = 4\,PD$
4. $PD = 15$

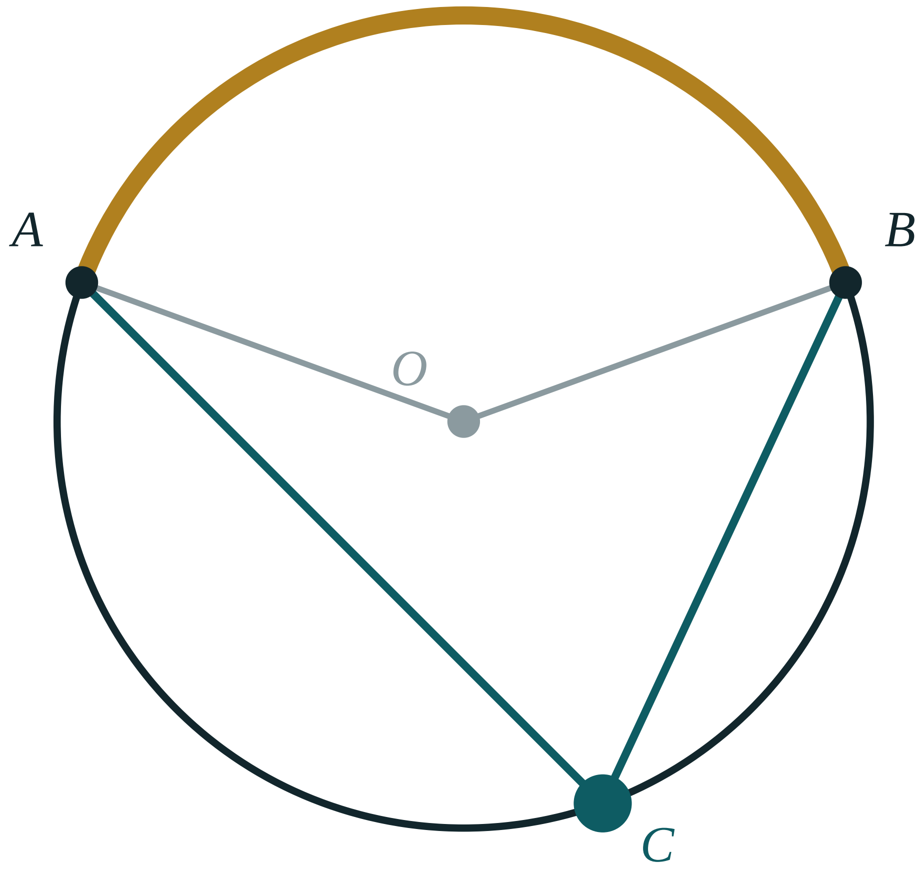
ANSWER

$PD = 15$

03 Interactive model

Draw one circle and three lines through a single external point; measuring the products with a ruler gives the same answer each time, and the theorem is discovered rather than told.

The power of a point



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Chord	Vatar	Хорда
Secant	Kesuvchi	Секущая
Tangent	Urinma	Касательная
Intersecting chords	Kesishuvchi vatarlar	Пересекающиеся хорды
External point	Tashqi nuqta	Внешняя точка
Power of a point	Nuqtaning darajasi	Степень точки
Product of segments	Kesmalar ko‘paytmasi	Произведение отрезков
Tangent–secant	Urinma-kesuvchi	Касательная–секущая

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Two chords meeting inside give:

- $PA + PB = PC + PD$
- $PA \cdot PB = PC \cdot PD$
- $PA = PC$
- nothing

For two secants, PB means:

- AB
- the whole secant from P
- half the secant
- the tangent

The tangent–secant theorem:

- $PT = PA \cdot PB$
- $PT^2 = PA \cdot PB$
- $PT = PA + PB$
- $PT^2 = PA + PB$

$PA = 4$, $PB = 9$: $PT =$

- 6
- 13
- 36
- 18

The power of a point on the circle is:

- R
- 0
- R^2
- undefined

A tangent is the limiting case of:

- a chord
- a secant
- a radius
- a diameter

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Chords: $PA = 4, PB = 9, PC = 6$: PD

   ANSWER 6

2. Chords: $PA = 3, PB = 8, PC = 4$: PD

   ANSWER 6

3. Tangent: $PA = 4, PB = 9$: PT

   ANSWER 6

4. Tangent: $PA = 2, PB = 8$: PT

   ANSWER 4

5. Secants: $PA = 5, PB = 12, PC = 4$: PD

   ANSWER 15

6. Power of a point on the circle

   ANSWER 0

7. PT^2 equals

   ANSWER $PA \cdot PB$

Medium · the standard question

7 PROBLEMS

1. Tangent with $PA = 4, AB = 5$: PT

   ANSWER 6

2. Tangent with $PT = 8, PA = 4$: PB

   ANSWER 16

3. Same: AB

   ANSWER 12

4. Tangent with $PT = 12, PB = 18$: PA

   ANSWER 8

5. Chords: $PA = 6, PB = 4$ and $PC = PD$: PC

   ANSWER $2\sqrt{6}$

6. Secants: $PA = 3, AB = 9, PC = 4$: PD

   ANSWER 9

7. Tangent with $PA = 5, AB = 11$: PT

   ANSWER $4\sqrt{5}$

Hard · stretch and reason

7 PROBLEMS

1. A chord of length 10 meets another at 4 from one end; the second is cut into 3 and x

   ANSWER $x = 8$

2. Two chords of a circle meet at the centre: the products

   ANSWER Both equal R^2

3. $OP = 13$, $R = 5$: the tangent length from P

   ANSWER 12

4. $OP = 13$, $R = 5$: the power of P

   ANSWER 144

5. A point 3 from the centre of a circle of radius 5: $PA \cdot PB$

   ANSWER 16

6. Two circles meet at A and B; P on line AB outside both

   ANSWER Equal tangent lengths to both

7. A tangent of length 15 and a secant with $AB = 16$: PA

   ANSWER 9

07 Homework

Homework — 5 tasks

For a secant, mark PA and PB from P on the figure before writing anything.

1. Two chords meet at P with $PA = 6$, $PB = 8$, $PC = 4$. Find PD.
2. A tangent and a secant from P give $PA = 3$ and $AB = 9$. Find PT.
3. Two secants give $PA = 4$, $PB = 15$ and $PC = 5$. Find PD.
4. A point is 10 from the centre of a circle of radius 6. Find the tangent length from it.
5. Prove the intersecting-chords theorem.

Practical exercises — constructions, scale drawing and similarity in use

Compasses, a scale and a measured field — geometry taken outside the exercise book.

LESSON 63–64 | 2 × 40 MIN | GEOMETRIYA 9, IV BOB AMALIY MASHQLARI | IGX 3.6

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The four constructions	13 min
Scale drawing	23 min
Measuring the unreachable	22 min
Loci	18 min
Homework	4 min

LEARNING OBJECTIVES

- Carry out the standard straight-edge and compasses constructions.
- Make and read a scale drawing, converting both ways.
- Use similar triangles to measure an inaccessible height or width.
- Construct a locus described in words.

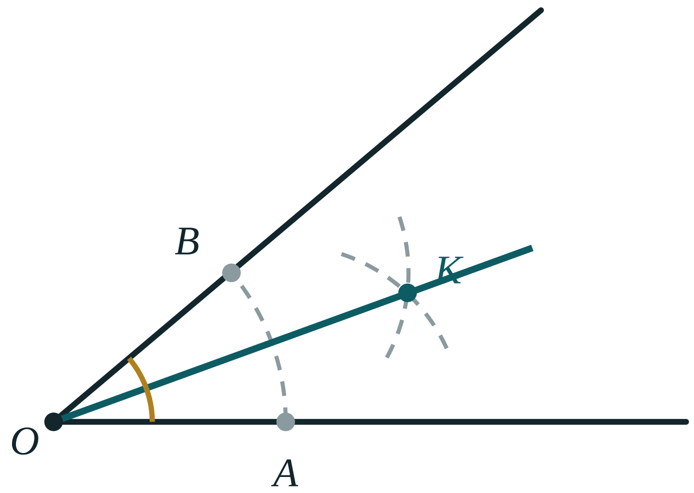
TEXTBOOK

National	pp. 217–220
Cambridge	Core & Extended, pp. 70–78
Workbook	Exercise 3.6

01 Explanation

The four constructions

CONSTRUCTION	METHOD	RESULT
perpendicular bisector of AB	arcs of equal radius from A and B	the set of points equidistant from A and B
bisector of an angle	arc across both arms, then two equal arcs	the set of points equidistant from the arms
perpendicular from a point to a line	an arc cutting the line twice, then a bisector	the shortest distance
an angle of $60°$	one arc, one equal step	an equilateral triangle



The angle bisector — one arc across both arms, then two equal arcs.

> **LEAVE THE CONSTRUCTION ARCS ON THE PAGE**
>
> Cambridge and the national papers both award marks for visible arcs. A perfect line with the working rubbed out scores less than a slightly wobbly one with the arcs showing.

Scale drawing

A scale of $1 : n$ means every length on the drawing represents n times as much in reality.

SCALE	1 CM REPRESENTS	1 CM² REPRESENTS
$1 : 100$	1 m	1 m²
$1 : 1000$	10 m	100 m²
$1 : 25\,000$	250 m	$62\,500$ m²
$1 : 50\,000$	500 m	$250\,000$ m²

> **AREAS SCALE BY N^2, NOT N**
>
> On a $1 : 1000$ plan, $1\ cm^2$ is $100\ m^2$ — because both dimensions are multiplied by 1000. This is the k^2 rule of Chapter I, in its most practical form.

Measuring the unreachable

TO MEASURE	METHOD
the height of a tree	compare its shadow with a metre rule's shadow
the height of a building	a mirror on the ground, similar triangles
the width of a river	a baseline on one bank and two measured angles
a distance across a lake	a triangle with two measured sides and the included angle

The mirror method. Place a mirror on the ground between you and the building. Step back until you see the top in it. Then

$$\frac{\text{your height}}{\text{your distance to the mirror}} = \frac{\text{the building's height}}{\text{its distance to the mirror}}$$

because the angle of incidence equals the angle of reflection, making the two triangles similar.

EVERY METHOD HERE IS TWO SIMILAR TRIANGLES

A shadow, a mirror, a sighting stick: in each case one triangle is small enough to measure and one is not, and they are similar. Chapter I, used outdoors.

Loci

A **locus** is the set of all points satisfying a condition.

CONDITION	LOCUS
at a fixed distance r from a point	a circle of radius r
equidistant from two points	the perpendicular bisector of the segment
equidistant from two intersecting lines	the pair of angle bisectors
at a fixed distance from a line	two parallels
seeing a segment at $90°$	the circle on it as diameter

A question asking for a region — "nearer to A than to B **and** within $5\ cm$ of C" — is answered by drawing both loci and shading the overlap.

DRAW EACH CONDITION SEPARATELY, THEN INTERSECT

Exactly the method used for systems of inequalities in the algebra course. Two conditions, two constructions, one shaded region.

02 Worked examples

EXAMPLE 1

A plan has scale $1 : 200$. A room measures $4\ cm$ by $3\ cm$ on the plan. Find its real dimensions and area.

1. $4\ cm \times 200 = 800\ cm = 8\ m$
2. $3\ cm \times 200 = 600\ cm = 6\ m$
3. Real area $= 48\ m^2$.
4. Check: $12\ cm^2 \times 200^2 = 480\ 000\ cm^2 = 48\ m^2$ ✓

ANSWER

$8\ m \times 6\ m = 48\ m^2$

EXAMPLE 2 **A *1.6 m* student stands *2 m* from a mirror and sees the top of a building *15 m* beyond it. Find the building's height.**

1. $$\frac{1.6}{2} = \frac{h}{15}$$
   Similar triangles.
2. $2h = 24$
3. $h = 12\ m$
4. The mirror method needs no instruments beyond a tape.

ANSWER

$12\ m$

EXAMPLE 3 **Describe the locus of points equidistant from two intersecting lines.**

1. A point equidistant from two lines lies on a bisector of the angle between them.
2. Two lines make two pairs of vertical angles.
3. So there are two bisectors.
4. The locus is the pair of angle bisectors — two perpendicular lines.

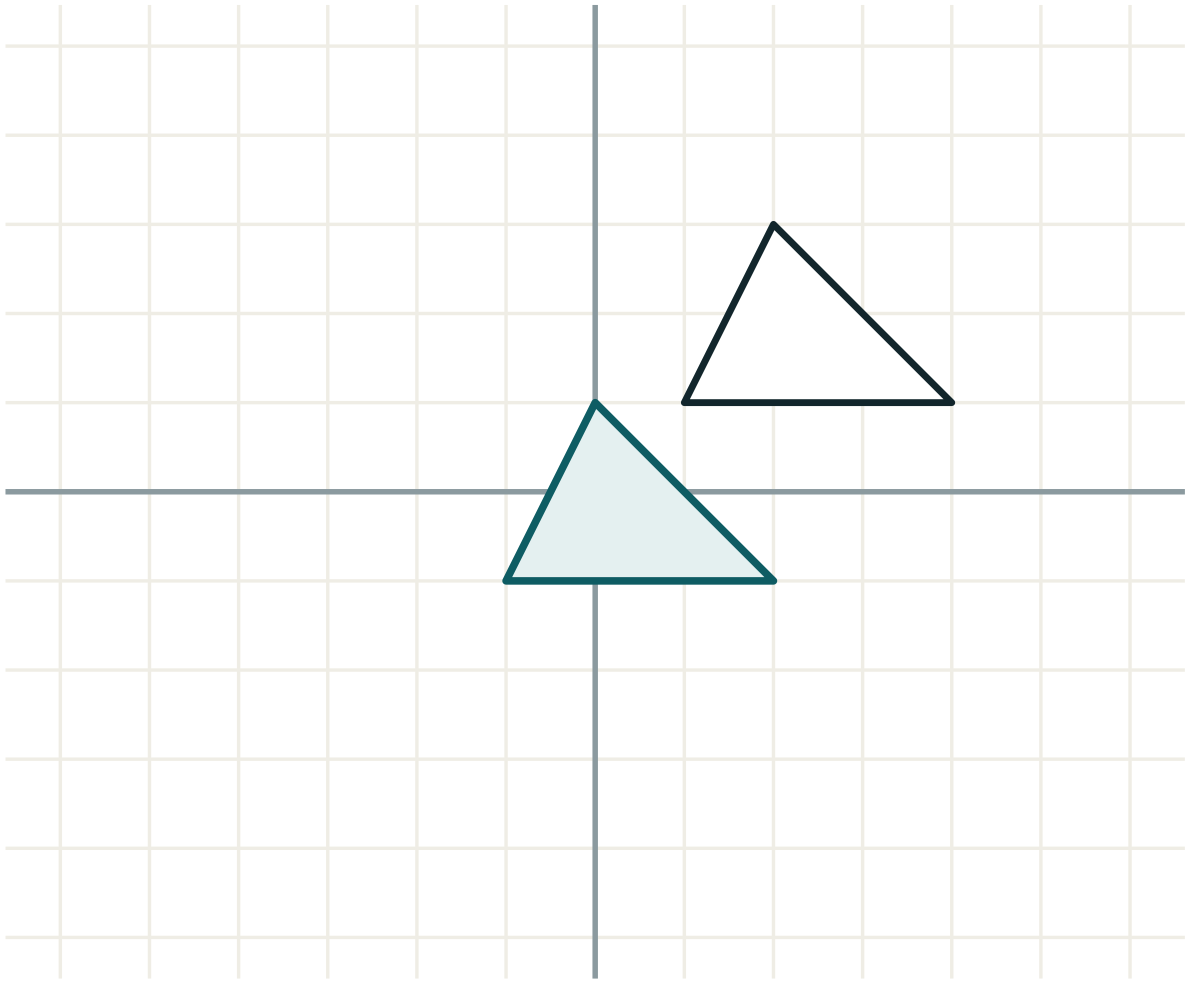
ANSWER

The two angle bisectors

03 Interactive model

Take the class outside with a tape and a mirror and measure the school building; the answer agrees with the plan on the wall, and the method is believed.

Scale and construction



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Scale drawing	Masshtabli chizma	Масштабный чертёж
Scale	Masshtab	Масштаб
Locus	Geometrik oʻrin	Геометрическое место точек
Equidistant	Teng uzoqlikda	Равноудалённый
Bisector	Bissektrisa	Биссектриса
Perpendicular from a point	Nuqtadan perpendikulyar	Перпендикуляр из точки
Inaccessible	Yetib boʻlmaydigan	Недоступный
Plan	Reja	План

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The locus of points equidistant from two points:

- a circle
- the perpendicular bisector
- two parallels
- an angle bisector

The locus at a fixed distance from a line:

- a circle
- one parallel
- two parallels
- a bisector

On a $1 : 1000$ plan, $1\ cm^2$ is:

- $10\ m^2$
- $100\ m^2$
- $1000\ m^2$
- $1\ m^2$

Construction arcs should be:

- rubbed out
- left visible
- drawn in pen
- omitted

The mirror method uses:

- Pythagoras
- similar triangles
- the cosine rule
- a protractor

The locus of points seeing a segment at $90°$:

- a line
- the circle on it as diameter
- two parallels
- a bisector

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Locus at distance r from a point

   ANSWER A circle

2. Locus equidistant from two points

   ANSWER The perpendicular bisector

3. Locus at distance d from a line

   ANSWER Two parallels

4. $1:100$: $1\ cm$ represents

   ANSWER $1\ m$

5. $1:1000$: $1\ cm$ represents

   ANSWER $10\ m$

6. $1:1000$: $1\ cm^2$ represents

   ANSWER $100\ m^2$

7. Construction arcs should be

   ANSWER Left visible

Medium · the standard question

7 PROBLEMS

1. $1 : 200$ plan, room 4×3 cm: real size

   ANSWER 8 $m \times 6$ m

2. Same: real area

   ANSWER 48 m^2

3. Mirror: 1.6 m at 2 m, building at 15 m

   ANSWER 12 m

4. Shadow: 1 m rule casts 1.5 m, tree casts 18 m

   ANSWER 12 m

5. $1 : 25\ 000$: 6 cm on the map

   ANSWER 1.5 km

6. Locus equidistant from two intersecting lines

   ANSWER The two angle bisectors

7. Locus seeing AB at $90°$

   ANSWER The circle on AB

Hard · stretch and reason

7 PROBLEMS

1. Points nearer to A than B and within $5\ cm$ of C

   ANSWER A region — half-plane ∩ disc

2. $1 : 50\ 000$: a lake of $4\ cm^2$ on the map

   ANSWER $1\ km^2$

3. A plan at $1 : 250$: a garden of $600\ m^2$

   ANSWER $96\ cm^2$ on the plan

4. Describe the locus of the centre of a circle of radius 2 rolling along a line

   ANSWER A parallel at distance 2

5. Describe the locus of points $3\ cm$ from a circle of radius $5\ cm$

   ANSWER Two concentric circles, radii 2 and 8

6. A field is surveyed as a triangle 80, 100, $120\ m$: its area

   ANSWER $\approx 3968\ m^2$

7. Its area on a $1 : 2000$ plan

   ANSWER $\approx 9.9\ cm^2$

07 Homework

Homework — 5 tasks

Leave every construction arc visible, and state the scale on every drawing.

1. Construct the perpendicular bisector of a $7\ cm$ segment, leaving all arcs.
2. Construct the bisector of a $70°$ angle.
3. A plan has scale $1 : 500$. A field is $6\ cm$ by $4\ cm$ on the plan. Find its real area.
4. A $1.7\ m$ person stands $2.5\ m$ from a mirror and sees the top of a mast $20\ m$ beyond it. Find the mast's height.

5. Describe and draw the locus of points equidistant from two parallel lines $6\ cm$ apart.

Control work 4 — proportional segments

Thales, the bisector, the altitude figure and the circle, in one short paper.

LESSON 65 | 1 × 40 MIN | GEOMETRIYA 9, NAZORAT ISHI 4 | IGX 11 REVIEW

LESSON CLOCK

2' | 25' | 8' | 5'

Instructions	2 min
The paper	25 min
Answers	8 min
Diagnosis and rewrite	5 min

LEARNING OBJECTIVES

- Apply Thales' theorem and the bisector theorem under time.
- Use the three relations of the altitude figure.
- Use the power of a point in all three configurations.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 183–220
Cambridge	Core & Extended, pp. 220–241
Workbook	Control paper G4

01 Explanation

The paper — 25 marks, 25 minutes

Q	TASK	MARKS	FROM
1	$DE \parallel BC$, $AD = 5$, $DB = 3$, $AE = 10$: find EC	5	L53–54
2	$AB = 12$, $AC = 20$, $BC = 24$: find BD for the bisector from A	5	L55–56
3	The altitude divides the hypotenuse into 5 and 20: find it and the legs	5	L57–58
4	Find the mean proportional between 6 and 24	4	L59–60
5	A tangent and a secant with $PA = 4$, $AB = 12$: find PT	6	L61–62

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the part-to-part ratio; Q2 two for $AB : AC = 3 : 5$; Q3 one for $AB = 25$; Q5 two for using $PB = 16$ rather than $AB = 12$.

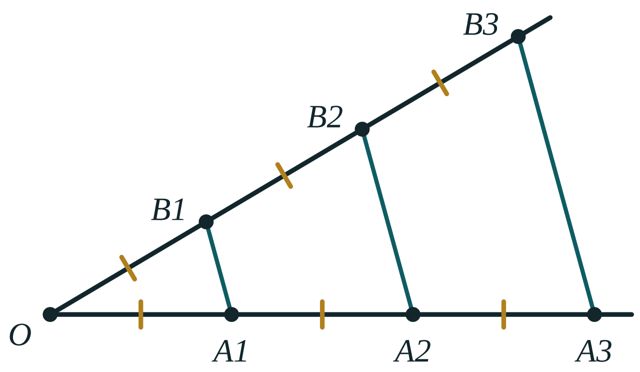
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
part and whole mixed	$\frac{AD}{AB} = \frac{AE}{EC}$	$\frac{AD}{DB} = \frac{AE}{EC}$
bisector ratio reversed	$\frac{BD}{DC} = \frac{AC}{AB}$	$= \frac{AB}{AC}$
leg paired with the wrong projection	$AC^2 = HB \cdot AB$	$AC^2 = AH \cdot AB$
hypotenuse taken as one part	$AB = 20$	$AB = 25$
mean proportional as the average	15	12
AB	$PT^2 = 4 \times 12$	$PT^2 = 4 \times 16$
square root omitted	$PT = 64$	$PT = 8$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter IV as one map

BLOCK	THE RELATION
Thales	$\frac{AD}{DB} = \frac{AE}{EC}$
part and whole	$\frac{AD}{AB} = \frac{DE}{BC}$
the bisector	$\frac{BD}{DC} = \frac{AB}{AC}$
the altitude	$CH^2 = AH \cdot HB$
each leg	$AC^2 = AH \cdot AB$
chords	$PA \cdot PB = PC \cdot PD$
tangent and secant	$PT^2 = PA \cdot PB$



Seven relations, all of them one proportion cross-multiplied.

EVERY LINE OF THAT TABLE IS $X^2 = AB$ OR $\frac{A}{B} = \frac{C}{D}$

There is only one idea in the chapter — a proportion between segments — and seven places where it appears. Recognising which configuration a question is in is the whole task.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: $AB = 12, AC = 20, BC = 24$.**

1. $\frac{BD}{DC} = \frac{12}{20} = \frac{3}{5}$
2. Total parts 8.
3. $BD = \frac{3}{8} \times 24 = 9$
4. $DC = 15$

   Check: $9 + 15 = 24$ ✓

ANSWER

$BD = 9$

EXAMPLE 2 **Model answer, Q3: projections *5* and *20*.**

1. $CH^2 = 5 \times 20 = 100 \Rightarrow CH = 10$
2. $AB = 25$

   The whole hypotenuse.
3. $AC^2 = 5 \times 25 = 125 \Rightarrow AC = 5\sqrt{5}$
4. $BC^2 = 20 \times 25 = 500 \Rightarrow BC = 10\sqrt{5}$

ANSWER

$CH = 10, AC = 5\sqrt{5}, BC = 10\sqrt{5}$

EXAMPLE 3 **Model answer, Q5: *PA = 4, AB = 12*.**

1. $PB = PA + AB = 16$

   The whole secant.
2. $PT^2 = PA \cdot PB = 4 \times 16$
3. $= 64$
4. $PT = 8$

ANSWER

$PT = 8$

03 Interactive model

Put four configurations on the board unlabelled and ask only which relation each needs; the recognition is what the paper tests.

Which relation?

Parallel lines cut two transversals: the segments are:

- equal
- proportional
- perpendicular
- unrelated

A line parallel to one side of a triangle divides the other two:

- equally
- in the same ratio
- perpendicularly
- in the ratio $2 : 1$

The bisector of an angle of a triangle divides the opposite side in the ratio of:

- the other two sides
- the other two angles
- $1 : 1$
- the altitudes

The altitude to the hypotenuse is the mean proportional between:

- the two legs
- the two segments of the hypotenuse
- the hypotenuse and a leg
- nothing

Each leg is the mean proportional between the hypotenuse and:

- the other leg
- its own projection on it
- the altitude
- the median

For a point outside a circle, the two secants satisfy:

- $PA \cdot PB = PC \cdot PD$
- $PA + PB = PC + PD$
- $PA = PC$
- nothing

For a tangent and a secant from the same point:

- $PT = PA$
- $PT^2 = PA \cdot PB$
- $PT^2 = PA + PB$
- $PT = PA \cdot PB$

Two chords meeting inside a circle give:

- $AE \cdot EB = CE \cdot ED$
- $AE + EB = CE + ED$
- $AE = CE$
- nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Thales’ theorem	Fales teoremasi	Теорема Фалеса
Bisector theorem	Bissektrisa teoremasi	Теорема о биссектрисе
Mean proportional	O‘rta proporsional	Среднее пропорциональное
Power of a point	Nuqtaning darajasi	Степень точки
Projection	Proyeksiya	Проекция
Tangent	Urinma	Касательная
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 uses:

- $\dfrac{AD}{AB} = \dfrac{AE}{AC}$
- $\dfrac{AD}{DB} = \dfrac{AE}{EC}$
- $AD = AE$
- the bisector theorem

In Q2 the ratio is:

- $AC : AB$
- $AB : AC$
- $AB : BC$
- $1 : 1$

In Q3 the hypotenuse is:

- 5
- 20
- 25
- 15

The mean proportional between 6 and 24:

- 12
- 15
- 18
- 144

In Q5, PB is:

- 12
- 16
- 4
- 8

The chapter's single idea is:

- area
- a proportion between segments
- congruence
- symmetry

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $AD = 5$, $DB = 3$, $AE = 10$: EC

   ANSWER 6

2. $AB = 12$, $AC = 20$: $BD : DC$

   ANSWER $3 : 5$

3. Same, $BC = 24$: BD

   ANSWER 9

4. Projections 5 and 20: the altitude

   ANSWER 10

5. Same: the hypotenuse

   ANSWER 25

6. Mean proportional of 6 and 24

   ANSWER 12

7. $PA = 4$, $AB = 12$: PB

   ANSWER 16

Medium · the standard question

7 PROBLEMS

1. $PA = 4, AB = 12$: PT

ANSWER 8

2. Projections 5 and 20: the legs

ANSWER $5\sqrt{5}$ and $10\sqrt{5}$

3. $AD = 6, DB = 4, AE = 9$: EC

ANSWER 6

4. $AB = 9, AC = 12, BC = 14$: BD

ANSWER 6

5. Chords $PA = 5, PB = 6, PC = 3$: PD

ANSWER 10

6. Mean proportional of 8 and 50

ANSWER 20

7. A tangent of 6 and $PA = 3$: PB

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. A right triangle with legs 15 and 20: the projections

   ANSWER 9 and 16

2. $DE \parallel BC$ with $[ADE] : [ABC] = 4 : 9$: $AD : DB$

   ANSWER $2 : 1$

3. A bisector cuts $BC = 35$ as $2 : 5$ with $AC = 25$: AB

   ANSWER 10

4. A tangent of $PT = 10$ and a secant with $AB = 21$: PA

   ANSWER 4

5. Two chords: $PA \cdot PB = 24$ and $PC = PD$: PC

   ANSWER $2\sqrt{6}$

6. A point P with $OP = 17$ and $R = 8$: the tangent length

   ANSWER 15

7. A rectangle 4×25: the side of the equal square

   ANSWER 10

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the annual revision.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 8$, $DB = 4$, $AE = 10$. Find EC.
2. A bisector from A divides $BC = 30$ with $AB = 14$ and $AC = 21$. Find BD.
3. The altitude divides a hypotenuse into 8 and 18. Find the altitude and the legs.
4. Find the mean proportional between 9 and 49.

5. A tangent and a secant give $PA = 5$ and $AB = 15$. Find PT.

Annual revision, and the Grade 10 preview

Four chapters, four pictures — and the plane geometry course closed before solid geometry opens.

LESSON 66–68 | 3 × 40 MIN | GEOMETRIYA 9, YILLIK TAKRORLASH | IGX 1–12 REVIEW

LESSON CLOCK

25' | 35' | 35' | 20' | 5

Four chapters, four pictures	25 min
Which chapter is this?	35 min
Mixed practice	35 min
What Grade 10 expects	20 min
The summer plan	5 min

LEARNING OBJECTIVES

- State the central idea of each of the four chapters in one sentence.
- Recognise which chapter a mixed question belongs to.
- Identify personally weak areas and plan summer practice.
- See what Grade 10 solid geometry will assume.

TEXTBOOK

National	pp. 1–220
Cambridge	Core & Extended, full course
Workbook	Revision exercises

01 Explanation

Four chapters, four pictures

CHAPTER	THE SENTENCE	THE PICTURE
I — similarity and transformations	same shape, all lengths × k, all areas × k^2	a figure and its scaled copy
II — trigonometry in any triangle	the sine rule for a matched pair, the cosine rule otherwise	a triangle with all six elements
III — regular polygons and the circle	n congruent triangles, and the circle as their limit	a circle with a polygon inside
IV — proportional segments	one proportion, seven configurations	a triangle cut by a parallel

FOUR PICTURES ARE ENOUGH

If those four can be drawn from memory with their labels, the year is secure. Every formula in the course can be rebuilt from one of them.

Which chapter is this?

QUESTION	CHAPTER	BECAUSE
“Two similar polygons have areas 27 and 48…”	I	k^2
“Find the largest angle of a triangle $4, 7, 9$”	II	SSS — the cosine rule
“A regular polygon has interior angle $150°$”	III	the exterior angle
“A tangent and a secant from P…”	IV	$PT^2 = PA \cdot PB$
“The area of a segment…”	III	sector − triangle
“The altitude to the hypotenuse…”	I or IV	similar triangles, or the mean proportional

TWO CHAPTERS CAN BOTH BE RIGHT

The last row is genuinely either: Chapter I proves the three triangles similar, Chapter IV names the relations that follow. Choose whichever route you can carry out fastest.

Mixed practice

TASK	ANSWER
Two similar triangles, areas 16 and 81: the ratio of perimeters	$4 : 9$
$a = 7, b = 8, C = 60°$: find c	$\sqrt{57} \approx 7.55$
A regular hexagon of side 5: its area	$\frac{75\sqrt{3}}{2}$
A sector $R = 6, \theta = 60°$	6π
A bisector with $AB = 9, AC = 12, BC = 21$: BD	9
Chords $PA = 3, PB = 8, PC = 4$: PD	6

MIXED PRACTICE TRAINS THE CHOICE

Twenty questions from one chapter train a method; twenty from four chapters train the recognition — and it is the recognition an examination measures.

What Grade 10 expects

Grade 10 geometry is **solid** geometry: points, lines and planes in space. What it assumes from this year:

GRADE 10 TOPIC	NEEDS, FROM GRADE 9
lines and planes in space	parallels, perpendiculars, projections
the angle between a line and a plane	projection, and right-triangle trigonometry
sections of a cube and a prism	similar triangles and Thales
areas of sections	the ratio k^2
volumes (Grade 11)	the ratio k^3
the surface of a cylinder and a cone	the circumference and the sector

Nothing in solid geometry is new mathematics: it is plane geometry applied inside a three-dimensional figure, one plane at a time.

THE SUMMER PLAN, IN ONE LINE

One page a week: the similarity ratios k, k^2, k^3; the sine and cosine rules with one worked triangle each; the circle formulae; the altitude figure with all three relations. Four sessions, and September starts from a full page.

02 Worked examples

EXAMPLE 1 **Two similar triangles have areas *16* and *81*. Find the ratio of their perimeters.**

1. $k^2 = \dfrac{81}{16}$

   Chapter I.

2. $k = \dfrac{9}{4}$

3. Perimeters scale by k.

4. $9 : 4$

ANSWER

$9 : 4$

EXAMPLE 2 **In $\triangle ABC$, $a = 7$, $b = 8$, $C = 60°$. Find c and the area.**

1. $c^2 = 49 + 64 - 2 \cdot 56 \cdot 0.5$

   Chapter II.

2. $= 113 - 56 = 57 \Rightarrow c = \sqrt{57} \approx 7.55$

3. $S = \frac{1}{2} \cdot 56 \cdot \sin 60°$

4. $= 14\sqrt{3} \approx 24.2$

ANSWER

$c \approx 7.55,\ S = 14\sqrt{3}$

EXAMPLE 3 **A regular hexagon has side 5. Find its area, and the area of its inscribed circle.**

1. $S = \dfrac{3\sqrt{3}}{2} \times 25 = \dfrac{75\sqrt{3}}{2} \approx 65.0$

   Chapter III.

2. $r = \dfrac{5\sqrt{3}}{2}$

3. $\pi r^2 = \pi \times \dfrac{75}{4}$

4. $= \dfrac{75\pi}{4} \approx 58.9$

ANSWER

$\dfrac{75\sqrt{3}}{2}$ and $\dfrac{75\pi}{4}$

03 Interactive model

Give twelve questions in random order and ask only for the chapter of each — no solving. Recognition, practised on its own, is what a mixed paper rewards.

Which chapter, and which relation?

Similar triangles have:

- equal sides
- equal angles and proportional sides
- equal areas
- equal perimeters

Their areas are in the ratio:

- k
- k^2
- k^3
- 1

The sine rule is:

- $\frac{a}{\sin A} = \frac{b}{\sin B}$
- $a^2 = b^2 + c^2 - 2bc \cos A$
- $S = \frac{1}{2} ab \sin C$
- $a + b > c$

The cosine rule finds a third side from:

- two angles
- two sides and the angle between them
- three angles
- one side

The area of a triangle from two sides and the included angle is:

- $\frac{1}{2} ab \sin C$
- $ab \sin C$
- $\frac{1}{2} ab \cos C$
- ab

The circumference of a circle of radius r is:

- πr^2
- $2\pi r$
- πr
- $4\pi r$

The area of a sector of angle α radians is:

- $\frac{1}{2} r^2\alpha$
- $r^2\alpha$
- $r\alpha$
- $\pi r^2\alpha$

An inscribed angle is:

- equal to the central angle
- half the central angle
- twice it
- a right angle always

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Similarity	O‘xshashlik	Подобие
Solving triangles	Uchburchaklarni yechish	Решение треугольников
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Proportional segments	Proporsional kesmalar	Пропорциональные отрезки
Solid geometry	Stereometriya	Стереометрия
Plane	Tekislik	Плоскость
Preparation	Tayyorgarlik	Подготовка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

"Areas *27* and *48*" belongs to:

- Chapter I
- Chapter II
- Chapter III
- Chapter IV

"Largest angle of *4*, *7*, *9*" belongs to:

- Chapter I
- Chapter II
- Chapter III
- Chapter IV

"Interior angle $150°$" belongs to:

- Chapter I
- Chapter II
- Chapter III
- Chapter IV

"A tangent and a secant" belongs to:

- Chapter I
- Chapter II
- Chapter III
- Chapter IV

Grade 10 geometry is:

- more plane geometry
- solid geometry
- coordinate geometry
- trigonometry

The most useful thing to revise is:

- the whole book
- the four pictures
- the definitions
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Areas 16 and 81: k

ANSWER $9:4$

2. $a = 7, b = 8, C = 60°$: c

ANSWER $\sqrt{57}$

3. Regular hexagon of side 5: its area

ANSWER $\frac{75\sqrt{3}}{2}$

4. Sector $R = 6, \theta = 60°$

ANSWER 6π

5. $AB = 9, AC = 12, BC = 21$: BD

ANSWER 9

6. Chords $PA = 3, PB = 8, PC = 4$: PD

ANSWER 6

7. Interior angle of a regular 12-gon

ANSWER $150°$

Medium · the standard question

7 PROBLEMS

1. Two similar solids, volumes 8 and 125: the ratio of areas

   ANSWER $4 : 25$

2. Sides $4, 7, 9$: the largest angle

   ANSWER $\approx 106.6°$

3. Its area

   ANSWER ≈ 13.4

4. A segment, $R = 6, \theta = 90°$

   ANSWER $9\pi - 18$

5. Projections 4 and 9: the altitude

   ANSWER 6

6. A tangent with $PA = 4, AB = 12$

   ANSWER $PT = 8$

7. A regular hexagon of side 5: its inscribed circle area

   ANSWER $\frac{75\pi}{4}$

Hard · stretch and reason

7 PROBLEMS

1. A triangle $13, 14, 15$: its area, inradius and circumradius

   ANSWER $84, 4, 8.125$

2. A regular hexagon and its circumscribed circle: the ratio of their areas

   ANSWER $3\sqrt{3} : 2\pi$

3. A right triangle with legs $9, 12$: the altitude and the two projections

   ANSWER 7.2; 5.4 and 9.6

4. Two similar triangles: perimeters 20 and 28, smaller area 25

   ANSWER 49

5. A ship: $15\ km$ on $070°$ then $20\ km$ on $160°$

   ANSWER $25\ km$

6. A sector of perimeter 30 and radius 10: its area

   ANSWER 50

7. A cyclic quadrilateral with three angles $80°, 95°, 100°$: the fourth

   ANSWER $85°$

07 Homework

Summer work — 5 tasks, one a week

Not five tasks for one evening: one page a week, spread across the summer.

1. Week 1: draw a figure and its enlargement, and write the three ratios k, k^2, k^3.
2. Week 2: solve one triangle by the sine rule and one by the cosine rule, in full.
3. Week 3: draw a circle with an inscribed hexagon and write every circle formula on it.
4. Week 4: draw the altitude figure and write all three relations, then do ten problems on them.
5. Week 5: do Control work 4 again from a blank page and compare.